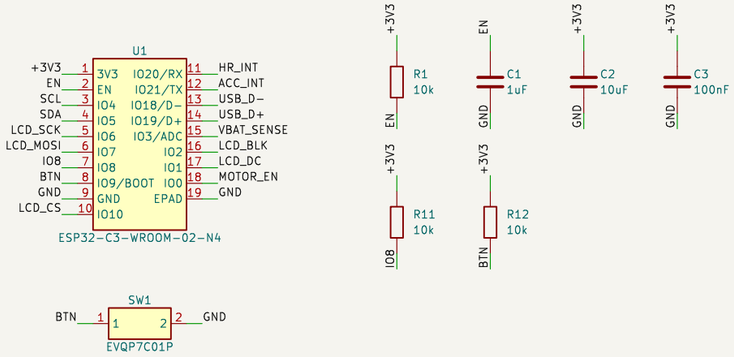
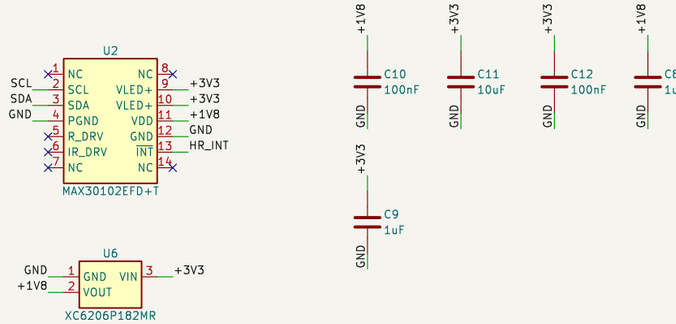
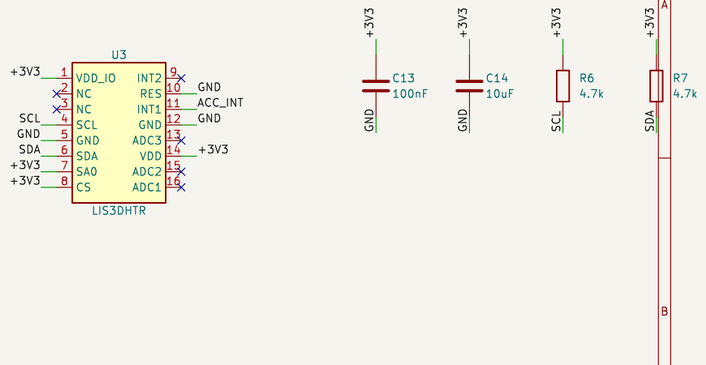
MCU ESP32-C3 (WiFi + BLE, native USB)



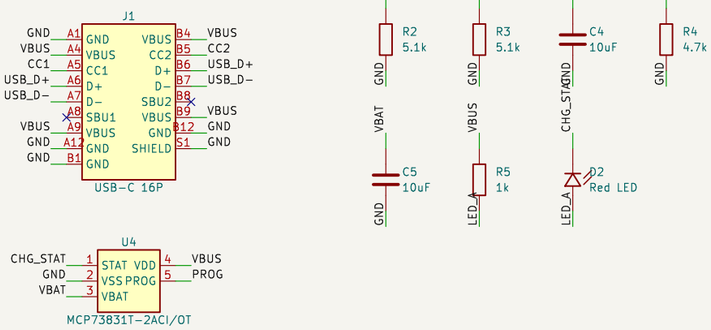
Heart rate + SpO2 (MAX30102, skin side)



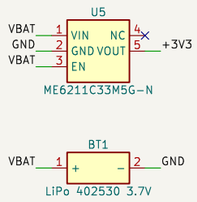
Step counter (LIS3DH)

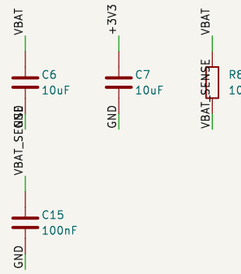


USB-C + Li-Po charger

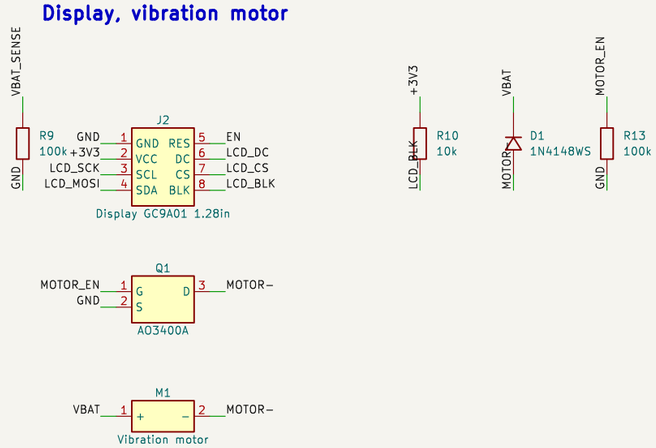


3.3 V supply + battery sense





Display, vibration motor



SPITALUL VIRTUAL smartwatch – made by Stancu Patrick

Every pin has a net label: same label text = same wire. Generated from scripts/design.py.

Power: USB-C 5V -> MCP73831 charger (~210 mA) -> Li-Po -> ME6211 3.3V. MAX30102 core runs from XC6206 1.8V.

Display: GC9A01 1.28in round 240x240 SPI, wired to the 8 edge pads. RES is tied to the ESP32 EN (resets together).

Hobby project – heart-rate / SpO2 readings are NOT for medical use.

ESP32-C3 + MAX30102 + LIS3DH, USB-C charging

made by Stancu Patrick

Sheet: /

File: spitalul_virtual_watch.kicad_sch

Title: Spitalul Virtual Smartwatch

Size: A3	Date: 2026-10-26	**Rev: 1.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1